Lecture 4: Probability

Lecturer

Tran Nguyen Lan

CLASS CODE

TOAE203

COURSE

Statistics and Probability in Economics and Business

Five parts, one question running through all of them

01

SAMPLE SPACE

An experiment defines a set of outcomes. Every probability is a share of that set

02

TWO EVENTS

Addition and multiplication rules; exclusivity and independence; the law of large number

03

CONDITIONAL PROBABILITY

Restricting the denominator, the base rate, and Bayes' theorem

04

PRACTICE

Three natural-frequency tables built by hand, then the same reasoning on VN-LISTED

SAMPLE SPACE

Event definition as a precondition for probability

Three defensible definitions of one phrase, applied to the same firms in the same year

Definition of "performs well"	Share of firms	Status of the figure
Return on assets above zero	___%	an empirical quantity
Return on assets above the industry median	___%	true by construction
Revenue growth above zero	___%	a different empirical quantity
A probability is a property of a defined experiment, not of a firm.		

What is the probability that a given listed firm performs well next year?

Write a number down before reading the table

SAMPLE SPACE

Experiment, sample space, event

Three objects defined in a fixed order; every probability presupposes all three

STEP 1

Experiment

A procedure with an uncertain outcome. Selected by the analyst, not by the data.

STEP 2

Sample space Ω (Không gian mẫu)

The set of all possible outcomes of that experiment.

STEP 3

Event

Any subset of Ω. Its probability is the share of Ω that it occupies.



Determine the Experiment, Sample Space and Event

Rolling a dice, 2 die

Dice Probability

Roll	Probability	Probability (%)
2	1/36	2.8%
3	2/36	5.6%
4	3/36	8.3%
5	4/36	11.1%
6	5/36	13.9%
7	6/36	16.7%
8	5/36	13.9%
9	4/36	11.1%
10	3/36	8.3%
11	2/36	5.6%
12	1/36	2.8%

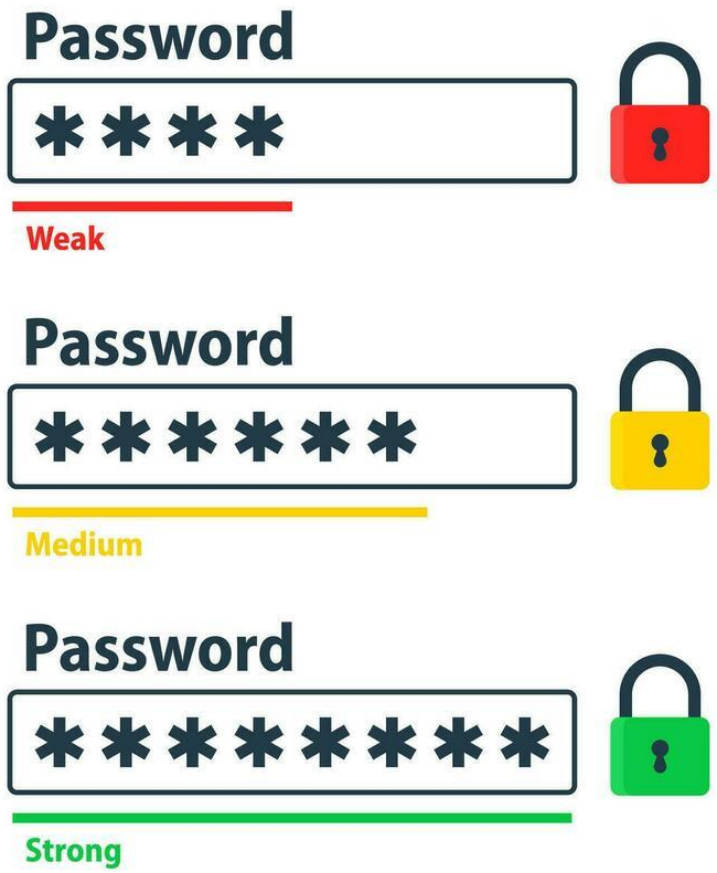
Determine the Experiment, Sample Space and Event

Roulette Wheel



Determine the Experiment, Sample Space and Event

Setting up your password



Time to Brute Force Crack a Password

Password	Characters	Time To Crack
apwmlq	6	19.5 Seconds
apwmlqr	7	8.33 Minutes
apwmlqrd	8	3.6 Hours
apwmlqrdc	9	3.91 Days
apwmlqrdcv	10	7.16 Years
apwmlqrdcvt	11	160 Years
apwmlqrdcvtz	12	9 Millenia

Source: Techferno

SAMPLE SPACE

Sample spaces generated by four experiments on one dataset

The data is unchanged in all four rows; only the experiment differs

Experiment	Ω consists of	Size of Ω
Draw one firm-year from the file	every firm-year record	___
Draw one firm from the latest year	firms present in that year	___
Draw one firm from Retail, latest year	Retail firms in that year	___
Draw one sample of 30 firms	every possible sample of size 30	___

Kolmogorov's axioms

Kolmogorov (1933), Grundbegriffe der Wahrscheinlichkeitsrechnung

THE AXIOMS

Three constraints define a probability measure on Ω

Every result in this session, including Bayes' theorem, is derived from these three statements:

- $0 \leq P(A) \leq 1$ — an event occupies a share of Ω
- $P(\Omega) = 1$ — some outcome in Ω occurs
- $P(A \cup B) = P(A) + P(B)$ for disjoint A, B

NOTATION

P(A) is an abbreviation, not a complete statement

The complete form is $P(A|B,S)$: Probability of event A to happen inside the experiment selected, on the set Ω defined for it).

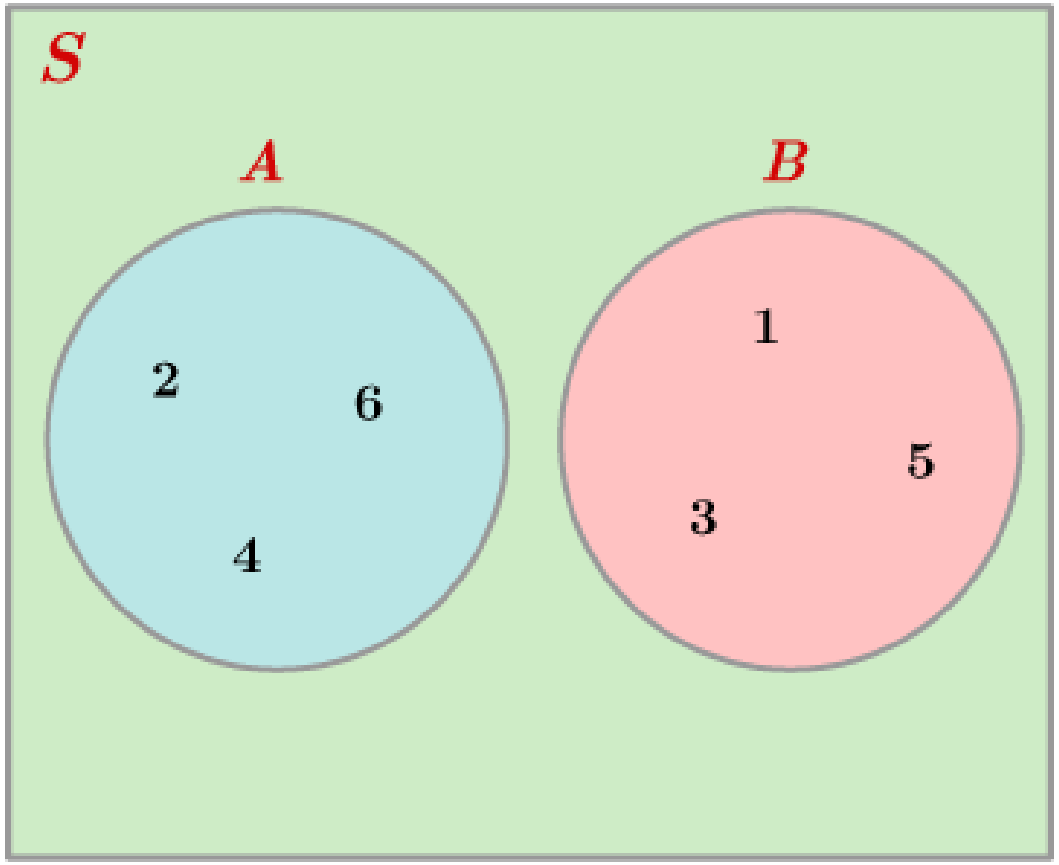
- The notation does not reveal the difference
- A reported probability is incomplete without the set it was computed over

The complement rule P(not A) = 1 – P(A) is a theorem, not a fourth axiom: it follows from the second and third.

Kolmogorov's axioms and the conditioning that notation suppresses

Kolmogorov (1933), Grundbegriffe der Wahrscheinlichkeitsrechnung

$$P(A \cup B) = P(A) + P(B)$$



Three interpretations of probability

Laplace (1812) · von Mises (1928) · de Finetti (1937)

Interpretation	The number rests on	Ω is specified by
Classical (cổ điển) - a priori	physical symmetry among outcomes	the symmetric outcomes
Frequentist (tần suất) - a posteriori	counting over a repeatable experiment	the experiment, which must be named
Subjective (chủ quan)	degree of belief, subject to coherence	no set is required
Frequentist is the major perspective in the Statistics course, but the Subjective is growing with the advances in technology		

Which of these three qualifies as a probability usable in analysis: a die showing six; 12% of listed firms reporting a loss; a 70% chance of a rate cut?

SAMPLE SPACE

Risk and uncertainty

Knight (1921), Risk, Uncertainty and Profit, Chapter VII

THE DISTINCTION

Two situations that standard usage conflates

Risk: the set of outcomes is known and their probabilities are known.

Uncertainty: one of those is unknown, or both.

- Risk — a roulette wheel; a bond class with a published default table
- Uncertainty — a new product; an unentered market; a regulation not yet drafted
- Most economically significant decisions fall in the second category

THE STATUS OF THE NUMBER

An assigned probability is a declarable assumption

Assigning a probability under uncertainty means treating the situation as a risk problem, and rests on a judgement about which prior cases are comparable to the present one.

- That judgement belongs to the analyst and is not contained in the data
- Same status as the bin width

The Addition rule

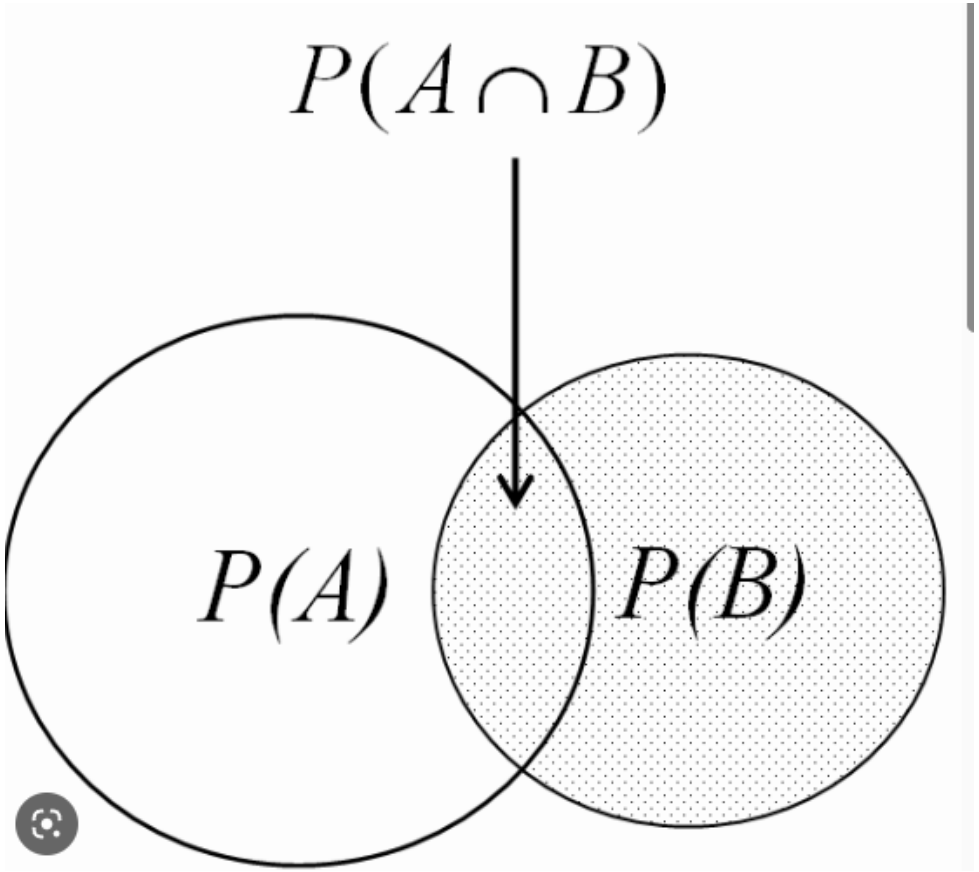
Derived from the third Kolmogorov axiom by decomposing A ∪ B into disjoint parts

THE ADDITION RULE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The subtracted term is the region counted twice: adding the two shares counts every case in the intersection once for A and again for B.

- The third axiom applies only to disjoint events
- For overlapping events the intersection must be removed once
- P(A) + P(B) is correct only when the intersection is empty



TWO EVENTS

The Multiplication Rule

The general form is stated first; the restricted form is derived from it

THE GENERAL RULE

$P(A \,\&\, B) = P(A) \cdot P(B \mid A)$

Valid without restriction. The probability of both events is the probability of A, multiplied by the probability of B conditional on A having occurred.

- Feller (1968), Vol. I, Ch. V treats this as the primitive form

THE RESTRICTED RULE

$P(A \,\&\, B) = P(A) \cdot P(B)$, only under independence

Independence is the condition $P(B \mid A) = P(B)$. Substituting that condition into the general rule produces the simpler expression.

How do you define 2 events are “independent”?

Independence ≠ Mutual exclusivity

One is a property of sets; the other is a property of information

Independent: the occurrence of one leaves the probability of the other unchanged.

$$P(A \mid B) = P(A) \quad \text{or} \quad P(B \mid A) = P(B)$$

The occurrence of A establishes with certainty that B did not occur.

Mutually exclusive: the intersection is empty.

$$A \cap B = \emptyset \Longrightarrow P(A \cap B) = 0$$

The occurrence of A establishes with certainty that B did not occur.

Two events can be exclusive, or independent, or neither — **but never both**

Classification of event pairs by exclusivity and independence

The two columns are filled from different evidence and do not covary

Event A	Event B	Exclusive	Independent
A firm-year is profitable	The same firm-year reports a loss	Yes	No
Number of customers on a random Friday	Total revenue on that same random Friday	No	No
First roll shows a six	Second roll shows a six	No	Yes
Today is rainy	The attendance rate is low	No	No (hopefully)
Independence is a hypothesis to be tested, not a simplification to be assumed.			

The law of large numbers: convergence in proportion, divergence in deviation

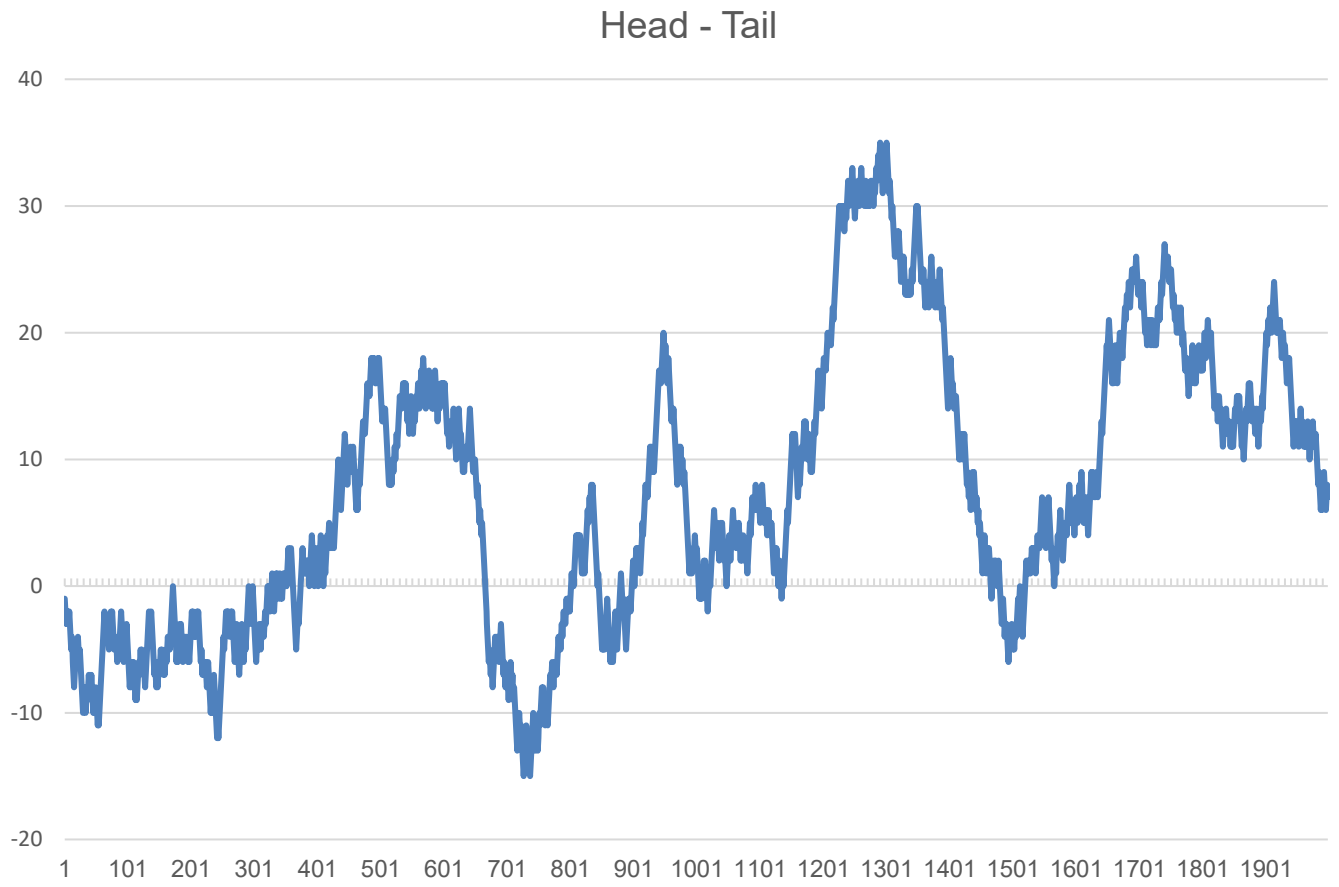
Jacob Bernoulli (1713), Ars Conjectandi, Part IV — the first proof, for Bernoulli trials

n	Expected |heads − tails|	Deviation as a share of n
100	≈ 8	≈ 8.0%
1,000	≈ 25	≈ 2.5%
10,000	≈ 80	≈ 0.8%
1,000,000	≈ 798	≈ 0.08%
The deviation grows with √n; its share of n falls with √n. Both hold at once.		

A fair coin is flipped n times. Is the deviation |heads − tails| larger at n = 100 or at n = 10,000?

The law of large numbers: convergence in proportion, divergence in deviation

Jacob Bernoulli (1713), Ars Conjectandi, Part IV — the first proof, for Bernoulli trials



Conditional probability

Probability of one event, given the other



[Monty Hall Simulator]

Conditional probability

Probability of one event, given the other

THE DEFINITION

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad \text{(with } P(B) > 0\text{)}$$

At first, the sample space is Ω.

Then, conditional event happens (B) discards every outcome outside the conditioning event, which then replaces Ω.

B becomes the new sample space

In this new space, the leftover of A is $P(A \mid B)$

THE CONSEQUENCE

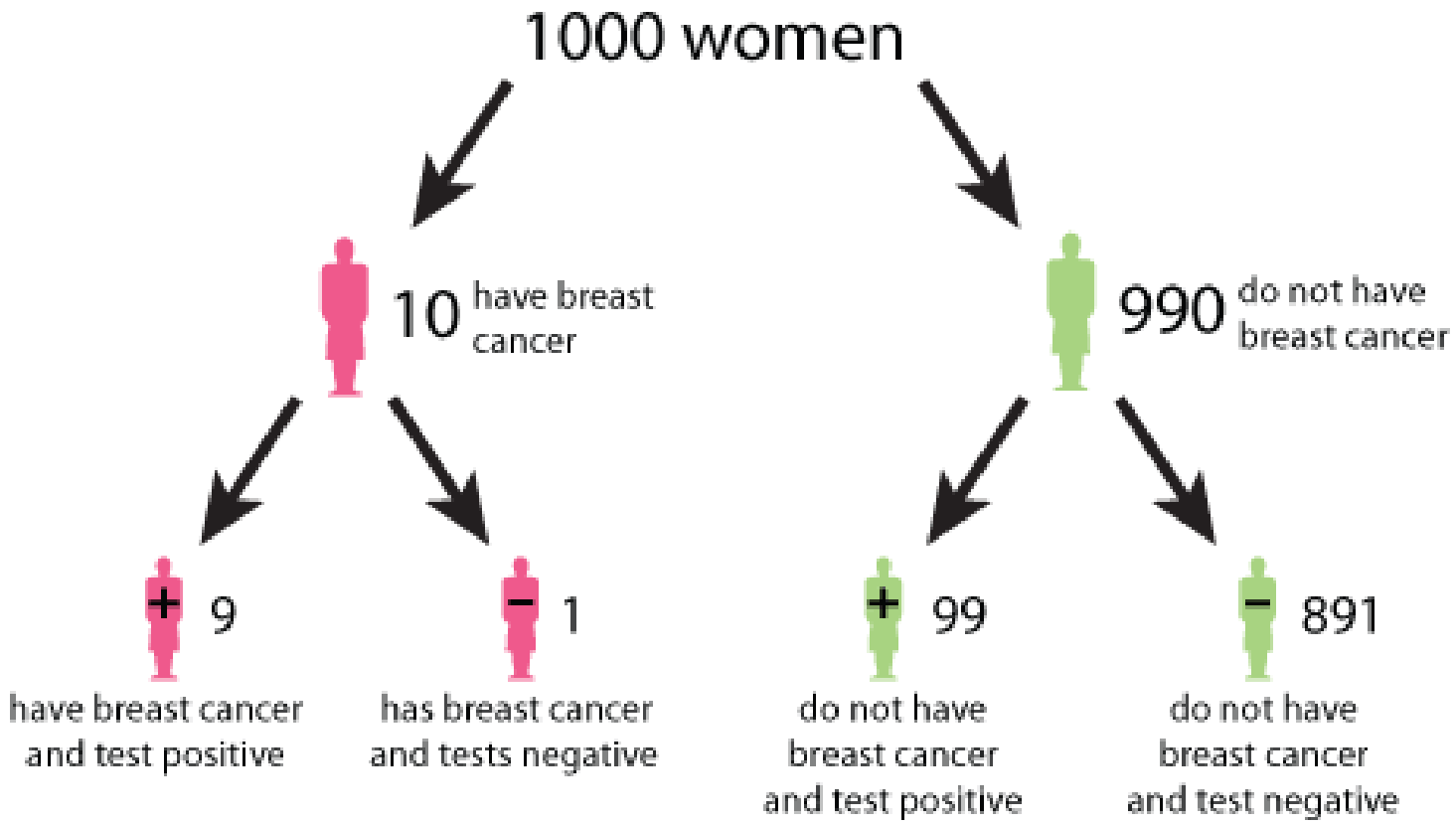
P(A|B) and P(B|A) are not the same

They are one numerator divided by two different denominators. No general relation constrains them to be close in value, and the remainder of this section establishes how far apart they can be.

- The ratio between them is P(A) / P(B)
- So they coincide only when A and B are equally probable

Diagnostic screening in natural frequencies

A cancer test proved to be effective at the rate of 90% — Gigerenzer & Hoffrage (1995)



An individual receives a positive result. What is the probability that this individual has the condition?

Diagnostic screening in natural frequencies

Base rate 1%, sensitivity 90%, false positive rate 10% — Gigerenzer & Hoffrage (1995)

Of 1,000 screened	Test positive	Test negative	Total
Has the condition	9	1	10
Does not have it	99	891	990
Total	108	892	1,000
P(condition | positive) = 9 / 108 = 8.33%			

One numerator, two denominators

The source of the discrepancy between 99% and 8.33%

THE TWO QUANTITIES

The value 9 is the same cell of the same table, twice

P(positive | condition) = 9 / 10, with the 10 who have the condition as denominator. P(condition | positive) = 9 / 108, with the 108 who tested positive.

- Numerator: identical
- Denominators: one is total of the condition, one is total of the state

INTERPRETATION

The 90% figure is correct and answers another question

The intuition producing the wrong answer omitted one quantity: the 9,900 individuals entering the screening already without the condition.

- Five percent of a large group exceeds ninety-nine percent of a small one
- A test reports P(evidence | state); a decision requires P(state | evidence)

The effect of the base rate on the posterior probability

Base rate fallacy — Tversky & Kahneman (1974); Bar-Hillel (1980), Acta Psychologica 44

Base rate	True positives	False positives	P(condition | positive)
1%	9	99	8.33%
10%	90	90	50%
50%	450	50	90%
Sensitivity 90% and false positive rate 10% are held constant in all three rows. The instrument does not change.			

Credit default screening: the same structure in a commercial setting

Base rate 2%, detection rate 90%, false alarm rate 8% on a book of 10,000 firms

Of 10,000 firms	Red flag	No flag	Total
Defaults	180	20	200
Does not default	784	9,016	9,800
Total	964	9,036	10,000
P(default | red flag) = 180 / 964 = 18.7%. Of the firms rejected, 81.3% would not have defaulted.			

A firm receives a red flag from the scoring model. What is the probability that it defaults?

The law of total probability

The name of the summation already performed in obtaining the denominator 594

STEP 1

Partition Ω

Split the sample space into cases that do and do not have the condition: 10 and 990.

STEP 2

Trace each route

Within each part, count the positives: 9 from the first, 99 from the second.

STEP 3

Sum the routes

P(positive) = 0.90 × 0.01 + 0.10 × 0.99 = 0.108 = 10.8%

A denominator is in general not a single quantity to be looked up. It is a sum over every route into the event.

Bayes' theorem

Bayes (1763), Phil. Trans. R. Soc. 53; general form independently by Laplace (1774)

THE EXPRESSION

$$\mathbf{P(A \mid B) \cdot P(B) = P(B \mid A) \cdot P(A)}$$

- P(A) is the prior (tiên nghiệm); P(A|B) is the posterior (hậu nghiệm)
- Bayes' theorem is an updating rule, not a position in the philosophy of statistics

WHAT IT ADDS

No information absent from the table

The advantage of the expression is compactness in written and computational settings. Every quantity in it was obtained by counting cases.

- The prior enters the calculation whether or not it is written down
- Setting the prior aside does not remove it; it sets it to an unstated value

The odds form of Bayes' theorem

Good (1950), Probability and the Weighing of Evidence

THE FORMULA

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

- Prior odds: 1 case with the condition for every 9 without it
- Likelihood ratio: how much more often a positive arises among those with the condition

WHY THIS FORM

Evidence multiplies; it does not add

A likelihood ratio of 19.8 is strong evidence. Applied to prior odds of 1 : 99 it still leaves the case improbable, because multiplication cannot overcome a sufficiently small starting ratio.

- The term odds (tỷ số cược) enters the course here
- Session 13 reports logistic regression coefficients as odds ratios
- Two independent pieces of evidence multiply their likelihood ratios

Transposed conditionals in published claims

Prosecutor's fallacy — Thompson & Schumann (1987), Law and Human Behavior 11(3)

Claim as stated	Quantity reported	Quantity inferred
A chance DNA match has probability one in a million, so the defendant is almost certainly guilty	P(evidence | innocent)	P(innocent | evidence)
90% of successful firms have a chief executive from a leading university	P(university | success)	P(success | university)
Most defaults exhibited warning indicator X beforehand	P(indicator | default)	P(default | indicator)
No claim contains an incorrect figure. The correction in every case is to state the denominator alongside the figure.		

The p-value as a conditional probability

ASA statement on p-values — Wasserstein & Lazar (2016), The American Statistician 70(2)

THE DEFINITION

P(data at least as extreme as observed | H_0 true)

The conditioning event is the truth of the null hypothesis. The p-value is a statement about the data, computed on the assumption that H_0 holds.

- It is a probability of data, not a probability of a hypothesis
- Goodman (2008) catalogues twelve distinct misreadings of it
- The ASA statement of 2016 lists this inversion first

THE MISREADING

P(H_0 true | data) is what it is taken to mean

The two differ exactly as the two quantities on slide 17 differ. Converting one into the other requires the prior probability of H_0, which the p-value does not contain.

- A small p-value shows the data would be improbable if H_0 held
- It does not show that H_0 is improbable

PRACTICE

Reconstructing the Breast Cancer case

Task A

Reduce the base rate tenfold

Base rate 0.1%, sensitivity 99%, false positive rate 5%. Use 100,000 so all counts are whole.

Task B

Improve the test, hold the base rate

Base rate 1%, sensitivity 99%, false positive rate cut from 5% to 1%. Use 10,000.

PRACTICE

Task C — Credit Default Case

A sentence of the kind that appears in credit and audit reporting

“80% of the defaults in this portfolio had been flagged by the model in advance. The model is therefore reliable.”

Name the conditional the sentence reports and the one the reader infers.

— Constructed for this exercise. It follows a pattern common in published risk reporting and describes no identified institution.

PRACTICE

The contingency table on VN-LISTED

Latest-year cross-section — the same slice used in Sessions 2 and 3

Build a PivotTable of industry × profitable in counts. Then compute the three probabilities as typed cell formulas, not a built-in.

TIME

12 minutes, following along in your own file

FIGURES

One table, three probabilities, and one independence check: P(A and B) against P(A) × P(B)

REQUIREMENTS

For each of the last two probabilities, point at the denominator cell and state whether it is a row total or a column total.

Recorded during the activity: "the numerator is cell ___; the denominators are ___ and ___."

PRACTICE

Row-wise and column-wise normalisation of one table

Two conditional distributions obtained from a single table of counts

Industry	Row-normalised: P(profitable | industry)	Column-normalised: P(industry | profitable)
Retail	___%	___%
Construction	___%	___%
Manufacturing	___%	___%
The second column is governed largely by industry size; the first is independent of it.		

Of P(profitable | Retail) and P(Retail | profitable), which is larger, and by roughly what factor?

PRACTICE

The sample space underlying these results

The experiment over which every figure in this section was computed

WHAT Ω WAS

Ω = the ___ listed firms in this file, latest year

Three nested sets are relevant and only the innermost is Ω: enterprises operating in Vietnam, firms listed on HOSE and HNX, and firms in this file with ten continuous years of data.

- Counts for the outer two sets are on the course reference sheet
- Those counts change each term and are deliberately not printed here
- Only the innermost set was actually drawn from

WHAT THIS PERMITS

Neither narrowing was a random draw

Listing requires a firm to be large enough and transparent enough to list. Ten continuous years of data requires a firm to have survived ten years.

- The figure is a correct statement about this Ω
- It is not yet a statement about Vietnamese enterprises generally
- Session 6 supplies the vocabulary and the machinery for that difference

Where a probability is reported, the sample space is reported with it.

SUMMARY

KEY TAKEAWAYS

1. PROBABILITY DEFINITION

2. SAMPLE SPACE AND EVENT

3. PROBABILITY OF 2 EVENTS

4. BAYESIAN PROBABILITY

5. LAW OF LARGE NUMBER

6. CONDITIONAL PROBABILITY